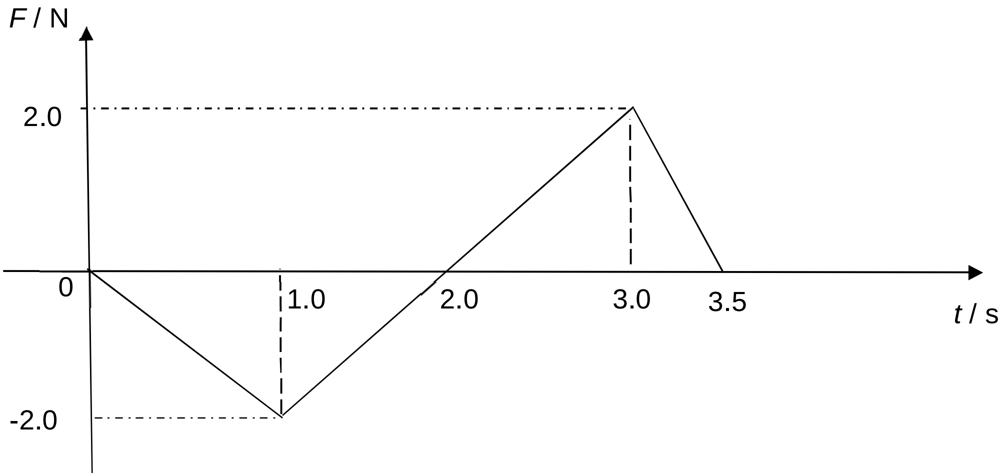
| | | |
|---|---|---|
| **5** | A resultant force $F$ acts on a mass of 150 g. The variation with time $t$ of $F$ is shown on the diagram below.   At time $t$ = 0 s, the mass is moving with a velocity of 2.0 m $s^{-1}$. What is the speed of the mass at $t$ = 3.0 s? | |
| | **A** | 2.7 m $s^{-1}$ |
| | **B** | 4.7 m $s^{-1}$ |
| | **C** | 6.7 m $s^{-1}$ |
| | **D** | 8.7 m $s^{-1}$ |

**Answer: B**

Change in momentum = area under the graph

= - ½ (2) (2) + ½ (2) (1)

= - 1.0 Nm

Therefore, m $\Delta v$ = - 1.0

$\Delta v$ = -1.0 / (150 x $10^{-3}$)

= - 6.67 m $s^{-1}$

Hence, $V_f - v_i$ = - 6.67

$V_f$ = -6.67 + 2.0 = - 4.67 m $s^{-1}$

Therefore the speed of mass = 4.7 m $s^{-1}$ (2 s.f.)